



SIH26089 — Cooperative Gig Services Platform for Household & Community Services

Problem Statement ID: 26089

Title: Cooperative Gig Services Platform for Household & Community Services

Ministry: Ministry of Cooperation

Department: National Council for Cooperative Training (NCCT)

Category: Software

Theme: Smart Automation

Prize: ₹1,00,000 INR

Deadline: 20 September 2026 (Portal: sih.gov.in) ^[1] ^[2]

Official Problem Statement (Full Description)

Background

Labour Cooperative Federations and Labour Cooperative Societies possess a large pool of skilled workers such as electricians, plumbers, carpenters, painters, domestic helpers, caregivers, drivers, gardeners, cleaners, and technicians. However, they lack a structured digital platform to connect these workers with households and institutions requiring such services. Private platforms currently dominate this market, while cooperative workers often remain underutilized despite having skills and local presence. ^[3] ^[4]

Problem Statement

To develop a **cooperative-owned digital service marketplace platform** that enables Labour Cooperative Federations and Labour Cooperative Societies to provide verified household and community services while ensuring **fair wages, worker welfare, and consumer trust**. ^[4] ^[5]

Expected Solution

Participants are expected to develop a comprehensive platform that includes: ^[6]

1. **Worker Profiles & Verification:** Digital profiles for cooperative workers with skill verification, ratings, and welfare benefit tracking.
2. **Service Matching & Booking:** AI-driven matching of service requests with available workers based on skills, location, and availability.
3. **Location-Based Recommendations:** Geo-fenced search and recommendations for nearby cooperative service providers.

4. **Fair Wage & Payment System:** Transparent pricing, cooperative commission structures, and direct payment settlement ensuring fair wages.
5. **Consumer Trust & Safety:** Verified workers, service guarantees, ratings/reviews, and grievance redressal.
6. **Dashboard for Cooperatives:** Admin dashboard for Labour Cooperative Federations to manage workers, track bookings, monitor welfare funds, and generate reports.
7. **Multilingual & Low-Literacy UX:** Voice-based booking, regional language support, and simple UI for workers and customers.

Detailed Research & Context

The Gig Economy in India

- India has an estimated 7.7 million gig workers today, projected to reach 23 million by 2030. ^[7]
- Gig workers are now formally recognized under Indian law as individuals earning from work arrangements outside traditional employer-employee relationships. ^{[8] [9]}
- The government is enhancing safeguards through portable social security benefits, a dedicated welfare fund, and a national registration framework. ^[9]
- The 2026 Periodic Labour Force Survey (PLFS) will for the first time track digital/platform workers experimentally. ^[10]

Cooperative Model Advantage

- **Bharat Taxi** (launched Feb 2026) is a government-backed cooperative platform where drivers are members/shareholders with voting rights, zero commission initially, and transparent surge-free pricing. ^[11]
- Cooperative platforms can bridge gaps by creating **worker-centric institutions** that engage with existing platforms while preserving local democratic governance. ^[7]
- Key benefits: fair wages, collective bargaining, transparent algorithms, social security, and profit-sharing among workers. ^{[11] [7]}

Existing Solutions & Gaps

- **Rozgar Setu** (research prototype): Digital platform for informal sector with authenticated profiles, skill-based matching, and location-aware search. Reduced worker discovery time by 80–90% in simulations. ^[12]
- **ShramSaathi:** Skill-based platform with video verification, location-based job discovery, and employer-worker chat. ^[13]
- **Gig Worker Discovery (Devpost 2026):** Two-sided platform with LLM-based semantic matching (not just keyword filtering). ^[14]

- **Gap:** No cooperative-owned, government-backed platform specifically for Labour Cooperative Federations to manage their own worker pool with welfare integration. ^[3] ^[4]

Unique Ideas & Differentiators

1. Cooperative Welfare Fund Integration

- Automatically deduct a small percentage from each transaction into a **worker welfare fund** managed by the cooperative.
- Funds can be used for health insurance, accident coverage, skill upgradation, and emergency aid (like Bharat Taxi's ₹5 lakh accident + family health insurance). ^[11]
- Transparent dashboard showing fund balance, contributions, and disbursements.

2. AI-Powered Skill Verification via Video

- Workers upload short **skill demonstration videos** (e.g., fixing a tap, wiring a switch).
- AI computer vision model validates skill proficiency and assigns **skill badges** (Beginner, Intermediate, Expert).
- Reduces fraud and builds consumer trust without expensive in-person verification.

3. Voice-First Booking for Low-Literacy Users

- Customers can book services via **voice notes** in regional languages (e.g., "Mujhe kal subah 10 baje plumber chahiye").
- NLP engine extracts: service type, location, time, urgency → auto-creates booking request.
- Workers receive voice notifications and can accept/decline via voice commands.

4. Dynamic Fair Pricing Algorithm

- ML model suggests **fair prices** based on:
 - Service type & complexity
 - Worker skill level & ratings
 - Local market rates (scraped from private platforms)
 - Distance & travel time
 - Urgency (same-day vs. scheduled)
- Ensures workers earn **20–30% more** than private platforms (which charge 20–30% commission). ^[11]

5. Cooperative Federation Dashboard

- Federations can:
 - Onboard new Labour Cooperative Societies
 - Monitor worker activity, earnings, and welfare contributions
 - Generate reports for government schemes (e.g., skill development grants)
 - Resolve disputes via integrated grievance cell
- API integration with NCCT for training program enrollment and certification.

6. Emergency SOS & Rapid Response

- One-click SOS button for workers in unsafe situations (e.g., hostile customer, accident on job).
- Alerts cooperative admin, nearby workers, and local police (via API integration).
- Live location tracking and incident logging for legal protection.

7. Offline-First Mobile App

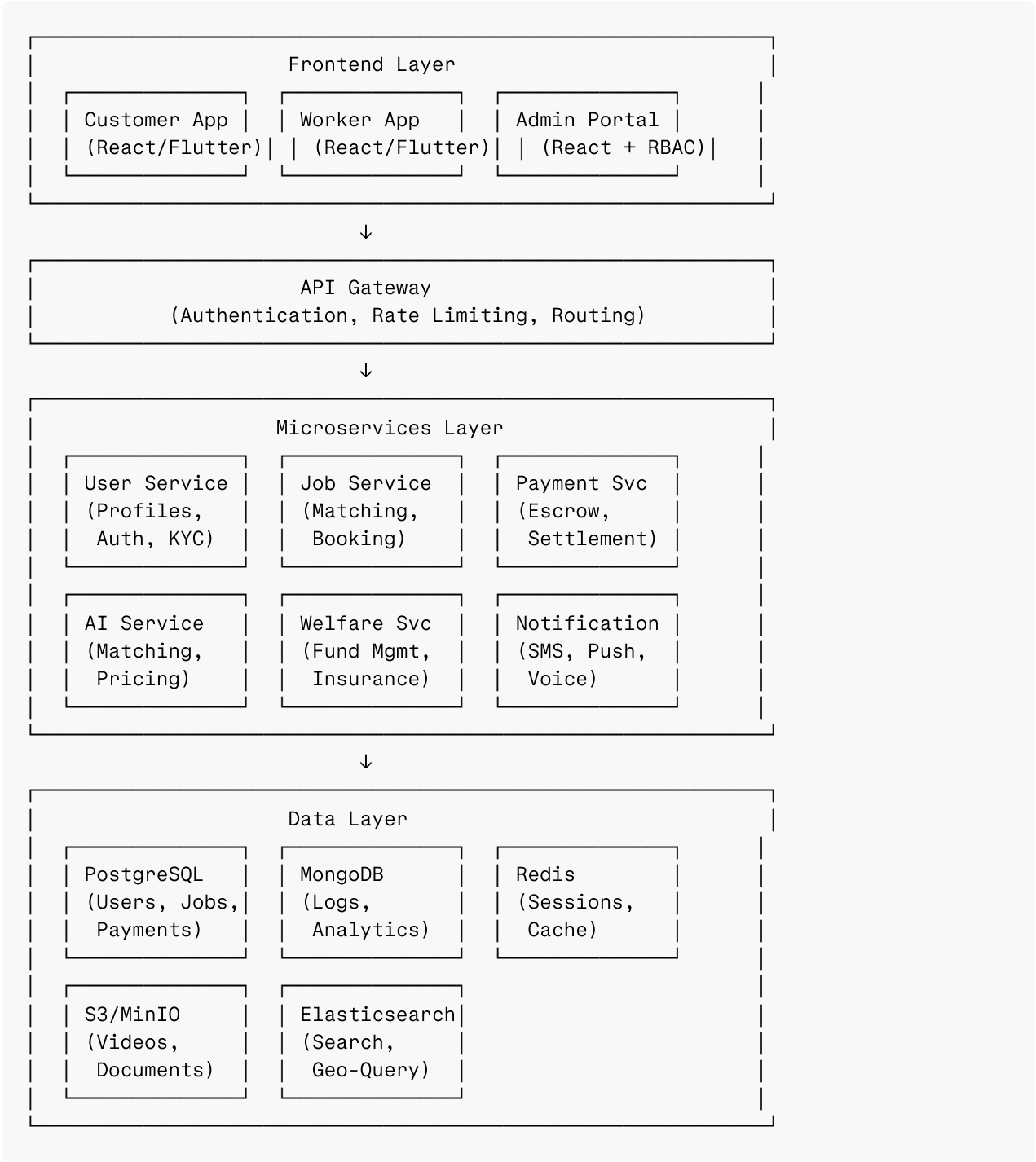
- Workers in low-connectivity areas can:
 - View assigned jobs offline
 - Mark job completion (syncs when online)
 - Access training videos (downloaded in advance)
- Ensures inclusivity for rural/semi-urban cooperative workers.

8. Gamified Skill Upgradation

- Workers earn points/badges for:
 - Completing jobs with high ratings
 - Attending cooperative training sessions
 - Referring new workers
- Points redeemable for:
 - Discounted toolkits
 - Priority job assignments
 - Welfare fund bonuses

System Design

High-Level Architecture



Tech Stack Recommendation

Component	Technology
Frontend (Mobile)	Flutter (cross-platform) or React Native
Frontend (Web Admin)	React.js + Tailwind CSS + ShadCN UI
Backend	Node.js (Express/NestJS) or Python (FastAPI)
Database	PostgreSQL (relational), MongoDB (logs/analytics)

Component	Technology
Caching	Redis (sessions, job queue)
Search & Geo	Elasticsearch (full-text + geo queries)
AI/ML	Python (FastAPI microservice), Hugging Face Transformers (NLP), YOLO/OpenCV (video verification)
Cloud	AWS/GCP/Azure (or NIC MeghRaj for govt compliance)
Authentication	JWT + OAuth 2.0 (Aadhaar e-KYC integration optional)
Payments	Razorpay/PhonePe API + Escrow logic
Notifications	Firebase Cloud Messaging (push), Twilio/Exotel (SMS/voice)
Video Storage	AWS S3 or MinIO (self-hosted)
DevOps	Docker, Kubernetes, GitHub Actions CI/CD

Database Schema (Simplified)

```
-- Users (workers, customers, admins)
CREATE TABLE users (
  id UUID PRIMARY KEY,
  name VARCHAR(255),
  phone VARCHAR(15),
  email VARCHAR(255),
  role ENUM('worker', 'customer', 'admin', 'cooperative'),
  cooperative_id UUID REFERENCES cooperatives(id),
  created_at TIMESTAMP
);

-- Worker Profiles
CREATE TABLE worker_profiles (
  user_id UUID PRIMARY KEY REFERENCES users(id),
  skills JSONB, -- [{skill: "plumbing", level: "expert", verified: true}]
  rating DECIMAL(3,2),
  total_jobs INT,
  welfare_balance DECIMAL(10,2),
  video_url VARCHAR(500),
  location GEOGRAPHY(POINT)
);

-- Jobs/Bookings
CREATE TABLE jobs (
  id UUID PRIMARY KEY,
  customer_id UUID REFERENCES users(id),
  worker_id UUID REFERENCES users(id),
  service_type VARCHAR(100),
  description TEXT,
  location GEOGRAPHY(POINT),
  scheduled_time TIMESTAMP,
  status ENUM('pending', 'accepted', 'in_progress', 'completed', 'cancelled'),
  price DECIMAL(10,2),
  welfare_contribution DECIMAL(10,2),
  created_at TIMESTAMP
```

```
);

-- Cooperative Federations
CREATE TABLE cooperatives (
  id UUID PRIMARY KEY,
  name VARCHAR(255),
  region VARCHAR(100),
  welfare_fund_balance DECIMAL(15,2),
  admin_user_id UUID REFERENCES users(id)
);

-- Reviews
CREATE TABLE reviews (
  id UUID PRIMARY KEY,
  job_id UUID REFERENCES jobs(id),
  rating INT,
  comment TEXT,
  created_at TIMESTAMP
);
```

AI/ML Components

1. Semantic Job Matching (LLM-based)

- Embed job descriptions and worker profiles using sentence transformers (e.g., all-MiniLM-L6-v2).
- Cosine similarity + LLM reranking for "why this match" explanations. ^[14]
- Example: Customer says "need someone who can fix AC and do electrical wiring" → matches workers with both skills even if profile says "HVAC technician + electrician".

2. Dynamic Pricing Model

- Regression model (XGBoost/LightGBM) trained on:
 - Historical job prices
 - Worker skill level
 - Distance
 - Time of day (peak/off-peak)
 - Urgency flag
- Output: Suggested price range (min, fair, max).

3. Video Skill Verification

- YOLOv8 or MediaPipe for action recognition in uploaded videos.
- Classify actions: "using screwdriver", "soldering wires", "fixing pipe joint".
- Confidence score → auto-verify skill or flag for manual review.

4. Voice NLP for Booking

- Whisper API (or open-source Whisper) for speech-to-text in Hindi/regional languages.
- Custom NER model to extract: service type, location, time, urgency.

- Example: "Kal subah 10 baje plumber chahiye" → {service: "plumbing", time: "next_day_10am", urgency: "normal"}.

Impact Goals

- **Economic Empowerment:** Provide cooperative workers with year-round digital income, reducing dependency on irregular daily-wage work.
- **Fair Wages:** Ensure workers retain 85–90% of earnings (vs. 60–70% on private platforms). ^[11]
- **Social Security:** Integrate accident insurance, health coverage, and retirement savings via welfare fund. ^{[9] [11]}
- **Digital Inclusion:** Voice-first, multilingual UX enables low-literacy workers to participate in the digital economy.
- **Trust & Safety:** Verified workers, transparent pricing, and SOS features build consumer confidence.
- **Scalability:** Platform can onboard 10,000+ workers across multiple states via Labour Cooperative Federations.

Prototype Feasibility & Selection Potential

Criteria	Rating	Justification
Feasibility	★★★★★	Uses mature tech (Flutter, FastAPI, PostgreSQL, open-source AI models). No hardware dependencies.
Innovation	★★★★★	Cooperative ownership + AI matching + welfare fund integration is unique.
Impact	★★★★★	Directly empowers marginalized workers, aligns with govt's cooperative mission. ^[11]
Scalability	★★★★★	Cloud-native architecture supports pan-India rollout via NCCT network.
Selection Potential	★★★★★	High relevance to Ministry of Cooperation's mandate; strong social impact narrative.

References & Resources

- **Official SIH Portal:** sih.gov.in
- **Problem Statement Repository:** [GitHub - SIH 2026 Problem Statements](#) ^{[15] [6] [3]}
- **Bharat Taxi (Cooperative Model):** [Circular Learnings](#) ^[11]
- **Gig Worker Policy:** [PIB Factsheet](#) ^[9]
- **Similar Platforms:** Rozgar Setu, ShramSaathi, Gig Worker Discovery ^{[12] [13] [14]}

This problem statement is **highly practical for CSE-AIML teams** with full-stack + AI/ML capabilities. The cooperative angle differentiates it from existing gig platforms, and the welfare fund integration aligns perfectly with the Ministry of Cooperation's vision. ^{[4] [3] [11]}

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2. https://www.blinknbuild.in/Assets/SIH_2026_All_226_Problem_Statements_Master_Catalogue.pdf
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26. <https://www.lowcode.agency/blog/flutterflow-gig-marketplace-platform>

Additional Features for SIH 26089

Cooperative Gig Services Platform for Household & Community Services

The following five features are proposed additions to the existing solution. They are designed to strengthen the cooperative-owned model while complementing the problem statement's existing features such as worker verification, AI matching, welfare integration, voice booking, dashboards, SOS and offline access.

9. Cooperative Worker Digital Passport

Each cooperative worker receives a portable, continuously updated digital professional profile containing verified skills, training and certifications, cooperative membership, work history, completed jobs, ratings, safety training and relevant welfare/benefit status. The passport gives the cooperative a trusted long-term record of worker capability instead of treating the worker only as a one-time booking resource.

Core functions

- Verified skills and skill levels
- Training and certification history
- Completed jobs and service experience
- Ratings and service record
- Cooperative membership and worker status
- QR-based profile/credential sharing (where appropriate)

Process flow

Worker → Verification/Training → Digital Passport → Customer/Cooperative

10. Community Group Booking & Job Pooling

The platform can combine multiple nearby service requests of a similar type into a coordinated group booking. For example, several households in the same locality may request plumbing work during a common time window. The system can identify compatible requests and create an efficient service batch, reducing unnecessary travel and improving worker utilisation.

Core functions

- Group nearby requests by location and service type
- Combine compatible time windows
- Assign suitable workers to the pooled jobs
- Show customers the grouped-service status
- Improve worker utilisation and reduce travel time

Process flow

Multiple households → Demand pooling → AI grouping → Cooperative assignment → Multiple jobs

11. Protected Cooperative Contribution & Worker Welfare Pool

The payment model is designed so that the worker does not face an additional arbitrary platform commission beyond the defined cooperative contribution. A 20% transaction-level cooperative contribution is transparently displayed before payment. Of this contribution, 5% is earmarked for platform/app running costs. The remaining 15% is maintained as a worker-welfare pool for insurance, emergency assistance and year-end cooperative surplus/benefit allocation according to approved cooperative rules.

Core functions

- 20% total cooperative contribution is shown transparently
- 5% allocated to app/platform running costs
- Remaining 15% allocated to worker welfare purposes
- Welfare uses can include insurance and emergency assistance
- Any year-end surplus is separately accounted for and distributed/allocated according to cooperative rules
- Worker sees gross service amount, contribution, welfare allocation and final payable amount

Process flow

Customer payment → Transparent 20% cooperative contribution → 5% app operations + 15% welfare pool → insurance/emergency/year-end surplus

12. Women-Safe Service Assignment & Female Worker Preference

Customers can request a female service professional for services where they prefer or require one. The cooperative can maintain a pool of verified female workers, while the matching system considers skill, availability, location and the customer's stated preference. This creates a practical women-safety option without restricting customers from using the wider worker network.

Core functions

- Female worker preference during booking
- Verified female-worker pool
- Skill and availability based assignment
- Optional preferred service window
- Worker safety and SOS mechanisms remain available
- Admin monitoring of women-worker assignments

Process flow

Customer preference → Eligible female workers → AI matching → Safe service assignment

13. AI Coordinated Multi-Service Project Booking

For larger household or institutional projects such as renovation, the customer can create one project request instead of making separate bookings. The customer selects or describes all required services—for example, electrical work, plumbing, painting and carpentry. AI then decomposes the project into service tasks, identifies suitable cooperative workers and creates a coordinated schedule so that dependent activities occur in the correct sequence.

Core functions

- Single project request for multiple service categories
- AI identifies required trades from the project description
- Assigns electricians, plumbers, painters, carpenters, etc.
- Creates a dependency-aware timetable
- Coordinates worker arrival and task completion
- Updates one overall project timeline and status

Process flow

Project request → AI task decomposition → Multi-worker matching → Dependency-aware schedule → Coordinated execution

Updated Feature Set — Prototype Prioritisation

Feature	Recommended prototype treatment
Cooperative Worker Digital Passport	Worker profile + verified skills + certificates + job history
Community Group Booking & Job Pooling	Show 3–5 nearby requests being combined into one service batch
Protected Cooperative Contribution & Worker Welfare Pool	Payment breakdown showing 20% contribution and its allocation
Women-Safe Service Assignment & Female Worker Preference	Booking toggle: "Prefer a female service professional"
AI Coordinated Multi-Service Project Booking	One renovation project → multiple trades → AI-generated schedule

Note on the welfare allocation: The 20% structure above is written exactly as a prototype rule: 5% for app/platform running costs and the remaining 15% for the welfare pool. The internal split of that 15% between insurance, emergency assistance and year-end surplus should be finalized by the cooperative's approved policy before implementation.

Revised Welfare Contribution Model

Updated structure for the Cooperative Welfare & Payment feature

The earlier Skill Development & Training allocation has been removed. Its 3% has been added to the Year-End Cooperative Surplus allocation. The total cooperative contribution remains exactly 20%.

Allocation	Percentage	Purpose
App & Technology Running Cost	5%	Hosting, servers, maintenance, security, notifications and technical support.
Worker Insurance & Social Security	6%	Insurance support and long-term worker welfare benefits.
Emergency Assistance Fund	4%	Financial assistance for eligible emergencies and urgent worker support.
Year-End Cooperative Surplus	5%	Year-end cooperative surplus retained/distributed according to cooperative rules
TOTAL	20%	Transparent Cooperative Contribution

Recommended Feature Wording

Transparent Cooperative Contribution: The worker's service earnings are not subject to an arbitrary platform commission. A clearly disclosed 20% cooperative contribution is allocated to app operations, worker insurance and social security, emergency assistance, and the cooperative's year-end surplus. This creates a transparent, worker-focused and sustainable model.

20% Flow

20% Cooperative Contribution → 5% App Running Cost + 6% Insurance & Social Security + 4% Emergency Assistance + 5% Year-End Cooperative Surplus